

TAGCL-DDI: Two-stage Augmentation-Driven Graph Contrastive Learning for Drug-Drug Interaction Prediction

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Abstract—Drug–drug interactions (DDIs) occur when the simultaneous administration of multiple drugs alters their pharmacological effects or causes adverse reactions. Existing DDIs prediction methods typically extract features of drug pairs from a DDI event graph and utilize a multilayer perceptron classifier to predict interactions. However, relying solely on the single DDI event graph limits the encoder’s ability to capture comprehensive drug features. To overcome this limitation, we propose TAGCL-DDI, a novel Two-stage Augmentation-driven Graph Contrastive Learning framework enhance drug feature representation. In the first stage, TAGCL-DDI employs the multi-head graph attention network to adaptively learn edge weights within the DDI event graph, and then generates multiple augmented graph views by mapping the graph into diverse subspaces through different encoders, enabling the capture of richer interaction information. In the second stage, node-level contrastive learning further refines feature representations across subspaces. Experimental results on two benchmark datasets show that TAGCL-DDI achieves state-of-the-art performance in DDIs prediction.

Index Terms—two-stage augmentation, graph contrastive learning, multi-head graph attention network, view generator

I. INTRODUCTION

Combination therapy carries the risk of DDIs [1]. Such interactions may alter the pharmacological effects of the drugs involved, leading to enhanced or reduced efficacy, and severe adverse effects or toxic reactions [2]. Consequently, the accurate prediction of DDIs has emerged as a issue in pharmacology, contributing to the advancement of drug discovery and the optimization of clinical treatment strategies.

Graph Neural Networks (GNNs) have been widely employed in DDIs prediction due to their strong capability to model unstructured data. Xu et al. [3] proposed the dual GNN architecture to capture both direct and indirect interactions between drug substructures at multiple scales. Wang et al. [4] applied GCNs to learn drug features and employed the Graph Attention Network to extract interaction information from DDI event graphs. Similarly, GMPNN-CS [5] and SSI-

DDI [6] directly applied GNNs to model interactions between drug molecular substructures for predicting potential DDIs.

Recently, graph contrastive learning (GCL) approaches [7] have gained considerable attention in DDIs prediction due to their superior performance. Xiong et al. [8] generated augmented views by randomly shuffling node features and edge relationships in the DDI event graph. The original view was treated as a positive sample, whereas the shuffled versions served as negative samples, enabling the model to capture implicit patterns in rare DDI events. Augmented views derived from different molecules were used as negative samples to learn discriminative molecular representations. MSHGCL [9] exploited the hierarchical nature of GNNs by extracting molecular substructures at various levels. In this framework, local substructures from shallow layers were treated as negative samples, and global substructures from deeper layers served as positive samples, enabling explicit modeling of relationships between molecular subcomponents. However, Tian et al. [10] and Zhang et al. [11] indicate that augmented views contribute unevenly to DDIs prediction. Variation in contribution levels among these views poses a fundamental challenge to direct view-level alignment and may ultimately impair overall prediction performance.

To address the aforementioned challenges, we propose a Two-stage Augmentation-driven Graph Contrastive Learning framework for DDIs prediction, named TAGCL-DDI. In the first stage, the multi-head graph attention network (Multi-head GAT) is employed to generate K learnable augmented views of the DDI event graph while preserving its original topological structure. Each view is assigned independent, adaptive edge weights to better characterize the interaction strength between drug nodes, thereby mitigating the limitations of relying solely on the original graph. Multiple Relational Graph Convolutional Networks (R-GCN) [12] with distinct parameters are further employed to project the DDI event graph into diverse feature subspaces. The model enhances

discriminative capacity across these subspaces by minimizing the neighborhood contrastive loss function. In the second stage, we introduce an automated contrastive learning strategy by employing the learnable view generator that applies random masking to node features to generate positive and negative samples, thereby further strengthening the representational power of the R-GCNs.

The contributions can be summarized as follows:

- We propose a novel two-stage augmentation framework, TAGCL-DDI. In the first stage, the neighborhood contrastive learning maps the DDI event graph into distinct subspaces to model the complex interaction relationships between drugs. In the second stage, the automated graph contrastive learning strategy constructs learnable node-level positive and negative samples, thereby optimizing the feature representations within each subspace.
- Neighborhood contrastive learning is applied to generate augmented views by leveraging the multi-head graph attention network that adaptively learns edge weights between drug nodes within the DDI event graph. Within two augmented views, drug nodes connected to the anchor via edges serve as positive samples, while unconnected ones act as negative samples. The approach enhances the diversity of feature subspaces and overcomes the limitations of relying on single DDI view.
- Automated graph contrastive learning employs a learnable view generator that randomly masks node features in augmented views, optimizing subspace representations by minimizing the contrastive loss.
- Extensive experiments on two benchmark datasets validate the effectiveness of the proposed two-stage TAGCL-DDI framework.

II. METHODS

A. Problem Definition

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$ denote a DDI event graph, where $\mathcal{V}$ represents the set of drug nodes, $\mathcal{E}$ denotes the set of edges between drugs, and $\mathcal{R}$ indicates the set of DDIs relations. Let $X = \{x_1, \dots, x_N\}$ denote the feature matrix of nodes, and $A \in \{0, 1\}^{N \times N}$ represent the adjacency matrix. Each node of $\mathcal{G}$ can be viewed as a molecular graph M_v of a drug compound. The objective is to develop the predictive model that infers the interaction type between any given pair of drugs.

B. Molecular Representation Learning

For each drug molecular graph M_v , TrimNet [13] is employed to extract molecular feature representations [14]. Specifically, during the message passing phase [15], the hidden states of nodes are updated through T iterations by aggregating features from neighboring nodes and bond attributes. In the readout phase, the readout function is applied to generate a graph-level feature representation of the molecular graph.

During the message passing phase, The feature update of node i is given by:

$$M(s_i, s_j, e_{ij}) = \left\|_{k=1}^K (\alpha_{ij}^k \odot W_s^k s_j \odot W_e^k e_{ij}) \right\|, \quad (1)$$

$$\alpha_{ij} = \text{Softmax}(\sigma(u^\top [W_s s_i \parallel W_e e_{ij} \parallel W_s s_j])), \quad (2)$$

where α_{ij}^k represents the attention weights in the model, σ is a nonlinear activation function of the network, and W_s , W_e and u are learnable parameters.

$$s_i^{(t+1)} = LN \left(GRU \left(s_i^{(t)}, \sum_{j \in \mathcal{N}_i} m_{ij}^{(t+1)} \right) \right), \quad (3)$$

$$m_{ij}^{(t+1)} = M(s_i^{(t)}, s_j^{(t)}, e_{ij}^{(t)}), \quad (4)$$

where $\mathcal{N}_i$ denotes the set of neighboring nodes of node i .

During the readout phase, an LSTM-based Set2Set network [16] is used to generate the graph-level representation X_v of the molecular graph.

C. TAGCL

A two-stage data augmentation-driven graph contrastive learning method (TAGCL-DDI) is proposed for learning drug feature representations from DDI event graphs $\mathcal{G}$. The first stage employs neighborhood contrastive learning to project the DDI event graph into multiple subspaces, thereby capturing diverse interaction patterns. Building on this, the second stage utilizes automated contrastive learning to further refine the feature representations of these DDI event graph subspaces, as shown in Fig. 1.

1) *Neighborhood Contrastive Learning*: The DDI event graph contains multiple drugs and their connecting edges, implicitly encoding rich interaction information. TAGCL-DDI addresses this by introducing the Multi-head GAT to adaptively learn edge weights in the DDI event graph, generating K augmented views $\tilde{\mathcal{G}}^{(k)}$. It employs R-GCNs as the encoder to map the DDI event graph into diverse subspaces, thereby capturing varied interaction patterns.

$$\tilde{A}_{ij}^{(k)} = \frac{e^{(LeakyReLU(\varphi^{(k)\top} [W^{(k)} \mathbf{x}_i \parallel W^{(k)} \mathbf{x}_j]))}}{\sum_{v_p \in \mathcal{N}_i \cup \{v_i\}} e^{(LeakyReLU(\varphi^{(k)\top} [W^{(k)} \mathbf{x}_i \parallel W^{(k)} \mathbf{x}_p]))}}, \quad (5)$$

where $W^{(k)}$ and $\varphi^{(k)}$ represents the trainable weight vector, and $v_p \in \mathcal{N}_i$ signifies the set of neighboring of node v_i . When $A_{ij} = 0$, $\tilde{A}_{ij}^{(k)}$ is explicitly set to 0.

Specifically, at the l -th layer of R-GCN, the feature representation $h_{v \in \tilde{\mathcal{G}}^{(k)}}^{(l)}$ of drug nodes is updated as follows,

$$h_{v \in \tilde{\mathcal{G}}^{(k)}}^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{N}_v^r} \frac{1}{|\mathcal{N}_v^r|} W_r^{(l)} h_{u \in \tilde{\mathcal{G}}^{(k)}}^{(l)} + W_o^{(l)} h_{v \in \tilde{\mathcal{G}}^{(k)}}^{(l)} \right), \quad (6)$$

where $W_r^{(l)}$ and $W_o^{(l)}$ denote learnable parameters, and $\mathcal{N}_v^r$ represents the set of neighboring nodes of vertex v under

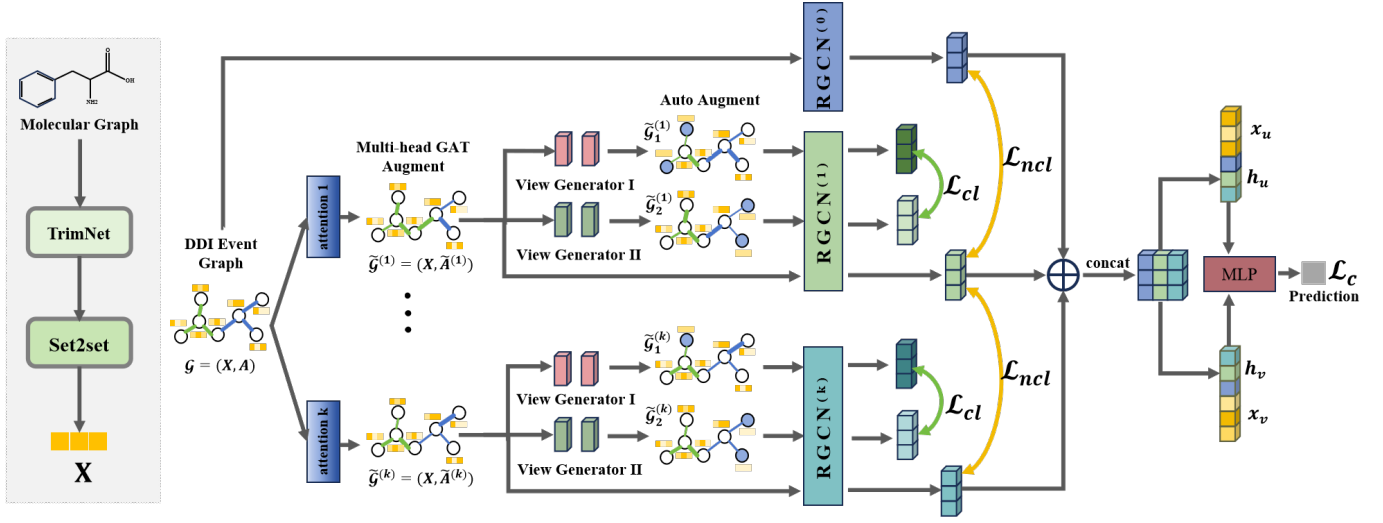


Fig. 1. The overall framework of TAGCL-DDI. The color of the edges in the DDI event graph represents the type of DDIs, and the thickness of the edges indicates the strength of the connection between nodes.

interaction type r . Finally, the drug features for the augmented view $\tilde{\mathcal{G}}^{(k)}$ are generated by integrating embeddings across all R-GCN layers.

$$h_{v \in \tilde{\mathcal{G}}^{(k)}} = \sum_{l=1}^L \alpha_l h_{v \in \tilde{\mathcal{G}}^{(k)}}^{(l)}, \quad (7)$$

where α_l represents trainable parameters that learn the contribution of the l -th layer's embeddings to the drug feature representation. For DDIs prediction, the drug features from the original DDI event graph are concatenated with those from each augmented view $h_v^{DDI} = h_{v \in \mathcal{G}} \| h_{v \in \tilde{\mathcal{G}}^{(1)}} \| \dots \| h_{v \in \tilde{\mathcal{G}}^{(k)}}$.

The DDI event graph $\mathcal{G}$ and the augmented view $\tilde{\mathcal{G}}^{(k)}$ share the same topological structure. Consequently, topological information is used as a supervised signal to construct positive and negative samples. For view $\tilde{\mathcal{G}}^{(1)}$ and view $\tilde{\mathcal{G}}^{(2)}$, we not only treat identical nodes across views as positive pairs $\{h_{v \in \tilde{\mathcal{G}}^{(1)}}^i, h_{v \in \tilde{\mathcal{G}}^{(2)}}^i\}$, but also the anchor node $h_{v \in \tilde{\mathcal{G}}^{(1)}}^i$ and its neighbors within one view, as well as the corresponding neighbors in the other view, as additional positive sample pairs $\{h_{v \in \tilde{\mathcal{G}}^{(1)}}^i, h_{v_j \in \tilde{\mathcal{G}}^{(1)}}^i \mid v_j \in \mathcal{N}_i\}$, $\{h_{v \in \tilde{\mathcal{G}}^{(1)}}^i, h_{v_j \in \tilde{\mathcal{G}}^{(2)}}^i \mid v_j \in \mathcal{N}_i\}$. All other node pairs are regarded as negative samples. Thus, the loss function for drug nodes consists of three components,

$$\mathcal{L}_{i,i}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(2)}) = -\log \frac{\exp(\text{sim}(h_{v \in \tilde{\mathcal{G}}^{(1)}}^i, h_{v \in \tilde{\mathcal{G}}^{(2)}}^i)/\tau)}{\theta_{\text{sim}}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(2)})}, \quad (8)$$

$$\mathcal{L}_{i,j \in \mathcal{N}_i}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(1)}) = -\log \frac{\exp(\text{sim}(h_{v \in \tilde{\mathcal{G}}^{(1)}}^i, h_{v \in \tilde{\mathcal{G}}^{(1)}}^j)/\tau)}{\theta_{\text{sim}}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(1)})}, \quad (9)$$

$$\mathcal{L}_{i,j \in \mathcal{N}_i}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(2)}) = -\log \frac{\exp(\text{sim}(h_{v \in \tilde{\mathcal{G}}^{(1)}}^i, h_{v \in \tilde{\mathcal{G}}^{(2)}}^j)/\tau)}{\theta_{\text{sim}}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(2)})}, \quad (10)$$

$$\begin{aligned} \theta_{\text{sim}}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(2)}) &= \sum_{i \neq n} \exp(\text{sim}(h_{v \in \tilde{\mathcal{G}}^{(1)}}^i, h_{v \in \tilde{\mathcal{G}}^{(1)}}^n)/\tau) \\ &+ \sum_n \exp(\text{sim}(h_{v \in \tilde{\mathcal{G}}^{(1)}}^i, h_{v \in \tilde{\mathcal{G}}^{(2)}}^n)/\tau), \end{aligned} \quad (11)$$

where $\mathcal{L}_{i,i}$ represents the loss for identical nodes across different views, $\mathcal{L}_{i,j \in \mathcal{N}_i}$ denotes the loss for neighboring nodes, $\text{sim}(\cdot)$ indicates the cosine similarity, and τ is the temperature parameter. Consequently, the objective function for the view is formulated as follows,

$$\begin{aligned} \mathcal{L}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(2)}) &= \frac{1}{N} \sum_{i=1}^K (\mathcal{L}_{i,i}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(2)}) \\ &+ \mathcal{L}_{i,j \in \mathcal{N}_i}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(1)}) \\ &+ \mathcal{L}_{i,j \in \mathcal{N}_i}(\tilde{\mathcal{G}}^{(1)}, \tilde{\mathcal{G}}^{(2)})). \end{aligned} \quad (12)$$

The overall objective function for neighbor contrastive learning is formulated as follows,

$$\mathcal{L}_{ncl} = \frac{1}{K} \sum_{k=1}^K \mathcal{L}(\mathcal{G}, \tilde{\mathcal{G}}^{(k)}). \quad (13)$$

2) Automated Contrastive Learning: During the neighbor contrastive learning phase, the DDI event graph is projected into distinct subspaces to extract drug feature representations $h_v^{DDI} = h_{v \in \mathcal{G}} \| h_{v \in \tilde{\mathcal{G}}^{(1)}} \| \dots \| h_{v \in \tilde{\mathcal{G}}^{(k)}}$. To this end, self-supervised contrastive learning is introduced to optimize the feature representations within each subspace.

As illustrated in Fig. 2, for each K -th augmented view, the GIN network [17] and the Gumbel-Softmax function [18] are employed as the view generator to select plausible node-level data augmentation operations,

$$f_v(\tilde{\mathcal{G}}^{(k)}) = \text{Gumbel} - \text{Softmax}(\text{GIN}(\mathbf{h}_{v \in \tilde{\mathcal{G}}^{(k)}})), \quad (14)$$

where f_v represents the data augmentation operation performed on node v , $f_v \in \{\text{Keep node}, \text{Mask node}\}$.

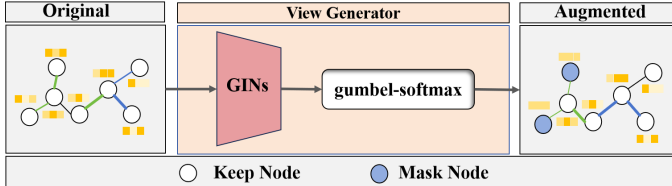


Fig. 2. Framework of the View Generator. Two types of node-level augmentation operations are applied: Keep Node retains the original node features without modification, while Mask Node replaces node features with the mean feature vector computed across all nodes, serving as the Mask value.

Two view generators with distinct parameters are constructed to generate two different views, denoted as $f_v^1(\tilde{\mathcal{G}}^{(k)})$ and $f_v^2(\tilde{\mathcal{G}}^{(k)})$. Subsequently, the parameter-shared R-GCN encoder is employed to extract node features from the augmented view $\tilde{\mathcal{G}}^{(k)}$ and the generated views, producing feature representations $z_{f_v^1}$ and $z_{f_v^2}$.

$$\mathcal{L}_{cl} = \frac{1}{N} \sum_{v \in \tilde{\mathcal{G}}^{(k)}} \left(1 - \frac{Z_{f_v^1} \cdot Z_{f_v^2}}{\|Z_{f_v^1}\|_2 \cdot \|Z_{f_v^2}\|_2} \right). \quad (15)$$

D. DDIs Prediction and Model Optimization

The drug molecular features x_v learned by trimNet are concatenated with the drug features h_v^{DDI} extracted from the DDI event graph to form the final feature representation $h_v = x_v \| h_v^{DDI}$ of drug v . Following the configuration of MRCGNN [8], MLP and the softmax function are employed to predict the interaction types between drug pairs.

$$y_{pred} = \text{softmax}(\text{MLP}(h_{(u,v)})), \quad (16)$$

where y_{pred} denotes the predicted values.

$$\mathcal{L}_c = \sum_{r \in R} y_{gt}^r \log y_{pred}^r, \quad (17)$$

where y_{pred}^r denotes the predicted probability of the drug pair interacting with type r , and y_{gt}^r represents the ground truth label. Consequently, the total loss function for training TAGCL-DDI is formulated as,

$$\mathcal{L} = \mathcal{L}_c + \alpha \mathcal{L}_{ncl} + \beta \mathcal{L}_{cl}, \quad (18)$$

where α and β are hyperparameters that balance the contributions of different loss components.

III. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we present the datasets, implementation details, and comparative experiments with state-of-the-art methods, followed by ablation and case studies to validate the effectiveness of TAGCL-DDI.

A. Experimental Setup

1) *Datasets*: The performance of TAGCL-DDI is evaluated on two benchmark datasets. The Deng dataset [19] contains 572 drugs, 65 types of DDIs, and 37,264 DDI instances. The Ryu dataset [20] consists of 1,706 drugs, and 86 DDI types with 191,570 DDI instances.

2) *Implementation Details*: In neighborhood contrastive learning, four augmented views are constructed and a single-layer Graph Attention Network with a hidden embedding dimension of 128 is employed. For self-supervised contrastive learning, a 4-layer GIN is adopted as the view generator, maintaining a hidden embedding dimension of 128. The encoder adopts a 2-layer RCGN architecture with a hidden dimension of 64.

3) *Baseline Methods*: R-GCN [12]: It treats the DDI event graph as a multi-relational graph, employing the R-GCN network to extract drug features from the DDI event graph. It feeds the jointly concatenated features of drug pairs into the classifier to predict potential DDI types.

SSI-DDI [6]: It investigates the significance of interactions between drug substructures for DDIs prediction. It utilizes the GAT network on drug molecular graphs to extract multi-scale drug substructures and combines an attention mechanism to predict DDI types.

MRCGNN [8]: It proposes negative counterpart augmentation strategy, leveraging contrastive learning to optimize the R-GCN encoder. DSN-DDI [21]: It introduces multi-view representation learning framework constructed from distinct drug molecular graphs and drug-pair bipartite graphs. The drug molecular graph extracts features of drug substructures, while the drug-pair bipartite graph enables message passing among multiple substructures.

DFPDDI [22]: It proposes edge-aware augmentation strategy to generate enhanced views, combining contrastive learning to focus on personalized drug features and differential drug-pair interaction information.

MOLGAECL [23]: It employs the subgraph removal strategy to perform data augmentation on unlabeled molecular graphs. It utilizes the Graph Isomorphism Network (GIN) [17] as the encoder to construct the graph autoencoder, incorporates NT-Xent [24] loss for contrastive learning pretraining.

B. Comparison with the baseline approach

Table I and Table II present the DDIs prediction performance of TAGCL-DDI compared with baseline methods on two benchmark datasets.

Compared with all baseline methods, TAGCL-DDI achieves the best Accuracy and attains optimal performance on both Macro-F1 and Macro-Pre, while securing the second-best

result on Macro-Rec. These findings demonstrate the effectiveness of TAGCL-DDI.

TABLE I
DDIs PREDICTION PERFORMANCE ON THE DENG DATASET

Methods	Accuracy	Macro-F1	Macro-Rec	Macro-Pre
R-GCN	0.8695	0.7026	0.6878	0.7500
SSI-DDI	0.7866	0.4216	0.3896	0.5139
MRCGNN	0.8979	0.7791	0.7688	0.8101
DSN-DDI	0.8324	0.7736	0.7426	0.8192
DFPDDI	0.9054	0.7803	0.7611	0.8249
MOLGAECCL	0.8925	0.7649	0.7543	0.8022
TAGCL-DDI	0.9065	0.7854	0.7628	0.8298

TABLE II
DDIs PREDICTION PERFORMANCE ON THE RYU DATASET

Methods	Accuracy	Macro-F1	Macro-Rec	Macro-Pre
R-GCN	0.9284	0.8487	0.8291	0.8881
SSI-DDI	0.9008	0.6663	0.6287	0.7507
MRCGNN	0.9567	0.8894	0.8727	0.9221
DSN-DDI	0.9478	0.8880	0.8698	0.9193
DFPDDI	0.9607	0.9096	0.9065	0.9327
MOLGAECCL	0.9301	0.8709	0.8504	0.8519
TAGCL-DDI	0.9636	0.9134	0.8985	0.9409

C. Ablation Studies

To analyze the contribution of specific modules in TAGCL-DDI to DDIs prediction performance, we developed three modified architectures:

- w/o Two-stage Augmentation: The R-GCN encoder was directly used to extract drug feature representations from the DDI event graph, which were then fed into an MLP to predict potential DDI types.
- w/o View-generator: Only the Multi-head GAT was employed to generate augmented views from the DDI event graph. The model predicted DDIs by jointly leveraging drug features learned from both the original DDI event graph and the augmented views, without optimizing the R-GCN encoder.
- w/o DDI Event Graph: The model relied on drug feature representations learned from multiple augmented views for DDIs prediction, completely removing the features extracted directly from the original DDI event graph.

The comparison of DDIs prediction performance between TAGCL-DDI and its three variants on the Deng dataset is shown in Fig. 3. The most significant performance decline occurs when removing the two-stage augmentation strategy (w/o Two-stage Augmentation), demonstrating the capability of TAGCL-DDI to effectively extract drug features from DDI event graphs. The second-largest performance drop is observed when eliminating R-GCN encoder optimization (w/o View-generator), indicating that mapping DDI event graphs to diverse subspaces via multi-head graph attention mechanisms can effectively capture rich interaction patterns. Compared to the w/o View-generator variant, TAGCL-DDI achieves optimal

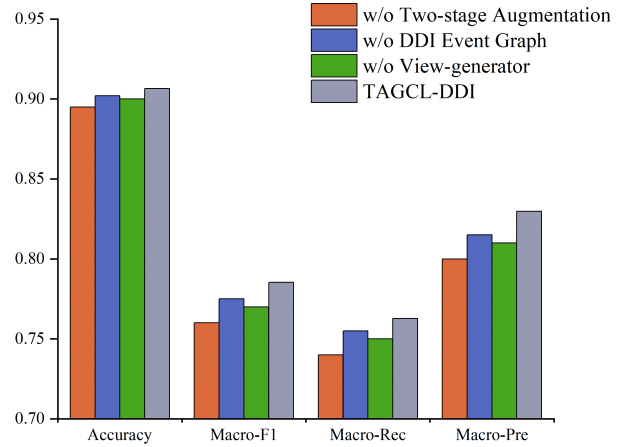


Fig. 3. Ablation Studies on Deng's Dataset

prediction performance, proving that training R-GCN encoders through self-supervised contrastive learning further optimizes subspace feature representations. Removing direct drug feature extraction from DDI event graphs (w/o DDI Event Graph) leads to performance degradation. This highlights that incorporating DDI event graphs provides essential complementary features for drug representation learning.

D. Hyperparameter Analysis

Effect of α and β . TAGCL-DDI is trained by jointly optimizing the classification loss, neighbor contrastive loss, and self-supervised contrastive loss. Therefore, analyzing the combined impact of α and β is essential. The DDIs prediction performance under various combinations of α and β is presented in Fig. 4. Given the different contributions of the two-stage contrastive learning components to DDIs prediction, the search space for α is set to $\{0.4, 0.5, 0.6, 0.7\}$, and for β to $\{0.0005, 0.005, 0.05, 0.1\}$. The results in Fig. 4 show that when $\alpha = 0.5$, both $\beta = 0.05$ and $\beta = 0.5$ achieve optimal Accuracy. However, $\beta = 0.05$ results in a lower Macro-F1 compared to $\beta = 0.5$. As a result, $\alpha = 0.5$ and $\beta = 0.5$ are selected as the optimal configuration to balance performance across all evaluation metrics.

E. Case Study

In the case study, Fig. 5 illustrates the interaction network among three commonly used drugs—Doxylamine, Eszopiclone, and Dofetilide—where solid and dashed lines represent validated and predicted DDIs, respectively. TAGCL-DDI successfully identifies both known synergistic and metabolic interactions as well as several novel, unverified interactions, offering valuable insights for future experimental validation.

IV. CONCLUSION

A two-stage augmentation-driven graph contrastive learning framework is proposed for DDIs prediction. In the first stage, TAGCL-DDI adaptively learns edge weights in the DDI event

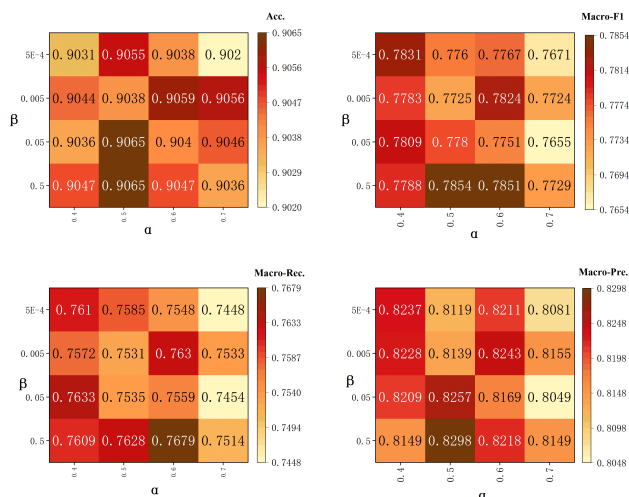


Fig. 4. The prediction performance of DDIs under the effect of α and β

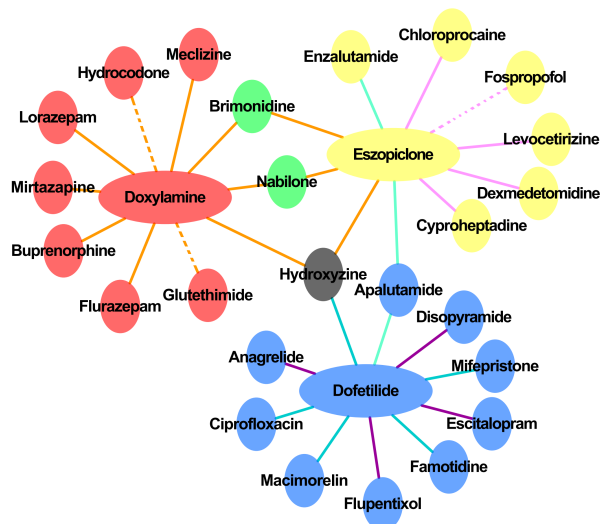


Fig. 5. Drug–drug Interaction Network

graph through the Multi-head-GAT, generating multiple augmented view-each corresponding to a distinct subspace of the original graph. By minimizing the neighborhood contrastive loss, TAGCL-DDI captures comprehensive drug interaction patterns. For each subspace, the view generator is introduced to establish node-level contrastive learning, in which the encoder is trained by minimizing the similarity loss to optimize feature representations within each subspace.

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